OSSP Skill Week 9

2520090058

Task 1-: To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings

```
  UW PICO 5.09                                  File: ..c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <signal.h>
#include <errno.h>

#define MAX_JOBS 20
#define MAX_CMD 256

/* Job states */
typedef enum {
    JOB_RUNNING,
    JOB_STOPPED,
    JOB_DONE,
    JOB_TERMINATED
} JobState;

/* Job record */
typedef struct {
    int job_id;
    pid_t pid;
    pid_t pgid;
    char command[MAX_CMD];
    JobState state;
} Job;

/* Global job table */
Job job_table[MAX_JOBS];
int job_count = 0;


/* Convert state to string */
const char *state_to_string(JobState state) {

    switch (state) {

        case JOB_RUNNING:
            return "Running";

        case JOB_STOPPED:
            return "Stopped";

        case JOB_DONE:
            return "Done";

        case JOB_TERMINATED:
            return "Terminated";

        default:
            return "Unknown";
    }
}


/* Add a job to the job table */
int add_job(pid_t pid, pid_t pgid, char *command) {

    if (job_count >= MAX_JOBS) {
        printf("Error: Job table is full.\n");
        return -1;
    }

    job_table[job_count].job_id = job_count + 1;

    job_table[job_count].pid = pid;

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text    ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text  ^T To Spell
```

OUTPUT:

```
                    ACTIVE JOBS
============================================================
JOB   PID         PGID        STATE           COMMAND
------------------------------------------------------------
2     1672        1672        Running         sleep 10
3     1673        1673        Running         sleep 15
============================================================

--- After 6 second(s) ---

============================================================
                    ACTIVE JOBS
============================================================
JOB   PID         PGID        STATE           COMMAND
------------------------------------------------------------
1     1672        1672        Running         sleep 10
2     1673        1673        Running         sleep 15
============================================================

--- After 7 second(s) ---

============================================================
                    ACTIVE JOBS
============================================================
JOB   PID         PGID        STATE           COMMAND
------------------------------------------------------------
1     1672        1672        Running         sleep 10
2     1673        1673        Running         sleep 15
============================================================

--- After 8 second(s) ---

============================================================
                    ACTIVE JOBS
============================================================
JOB   PID         PGID        STATE           COMMAND
------------------------------------------------------------
1     1672        1672        Running         sleep 10
2     1673        1673        Running         sleep 15
============================================================

--- After 9 second(s) ---

============================================================
                    ACTIVE JOBS
============================================================
JOB   PID         PGID        STATE           COMMAND
------------------------------------------------------------
1     1672        1672        Running         sleep 10
2     1673        1673        Running         sleep 15
============================================================

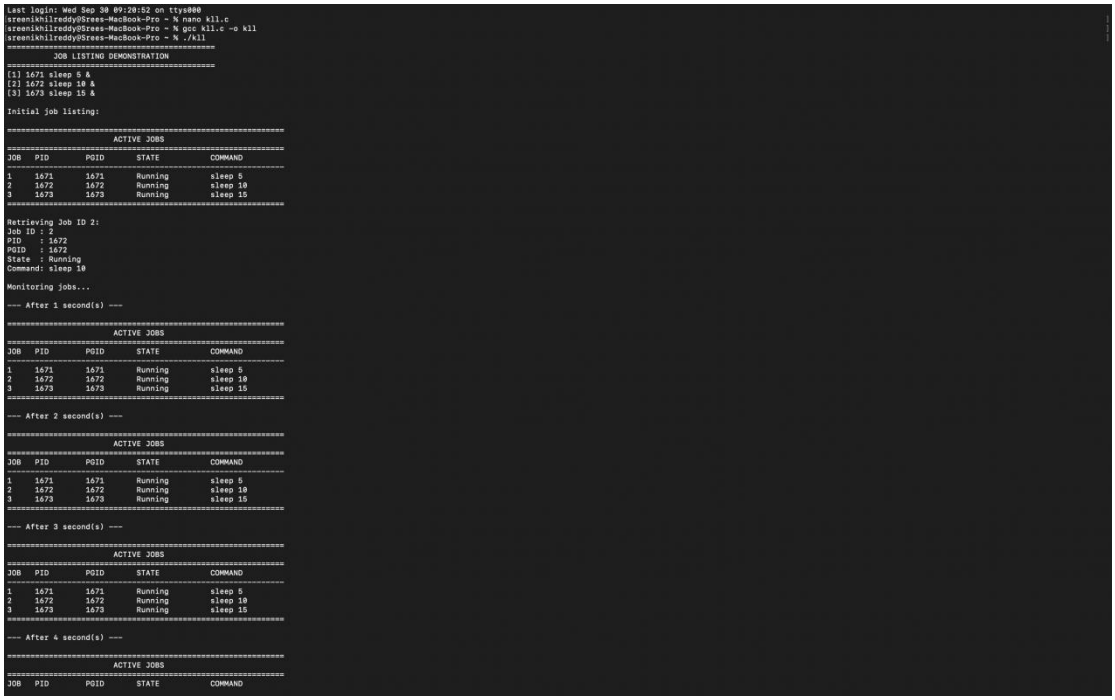
--- After 10 second(s) ---

============================================================
                    ACTIVE JOBS
============================================================
JOB   PID         PGID        STATE           COMMAND
------------------------------------------------------------
2     1673        1673        Running         sleep 15
============================================================

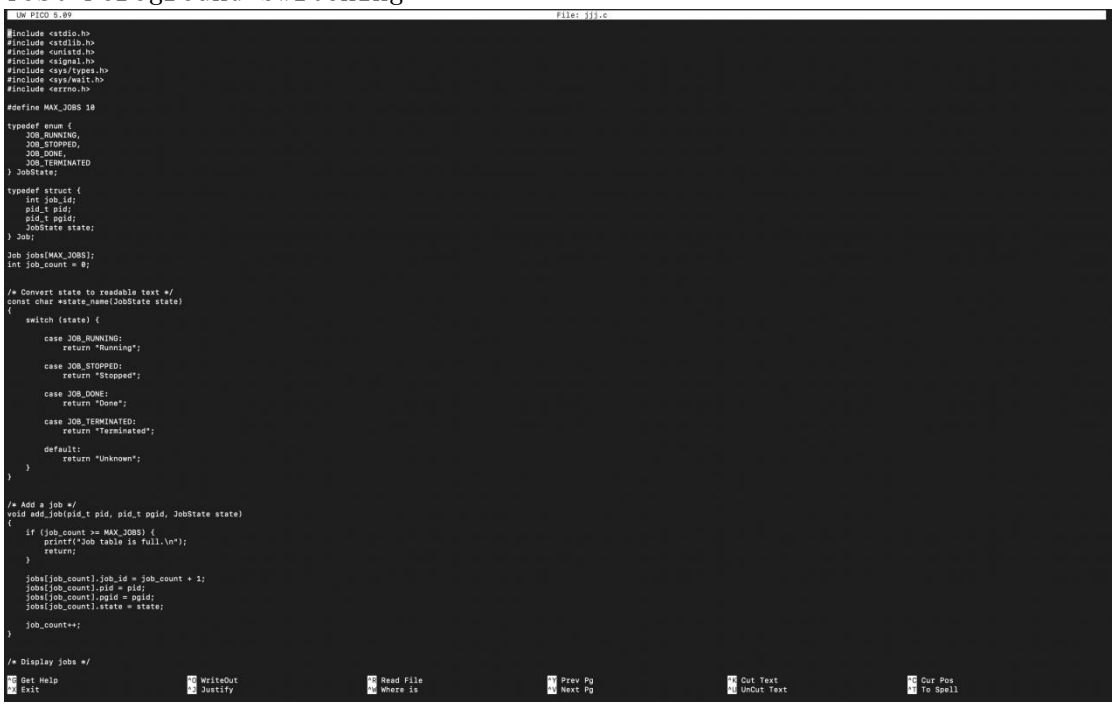
--- After 11 second(s) ---

============================================================
                    ACTIVE JOBS
============================================================
JOB   PID         PGID        STATE           COMMAND
------------------------------------------------------------
1     1673        1673        Running         sleep 15
============================================================

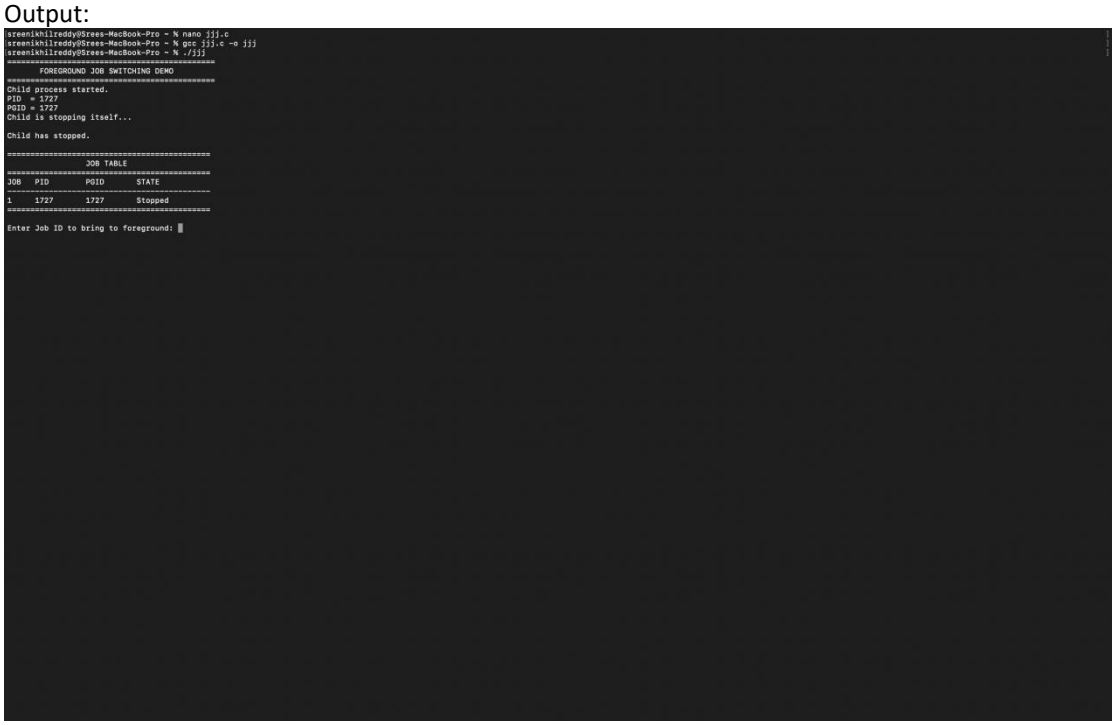
--- After 12 second(s) ---
```



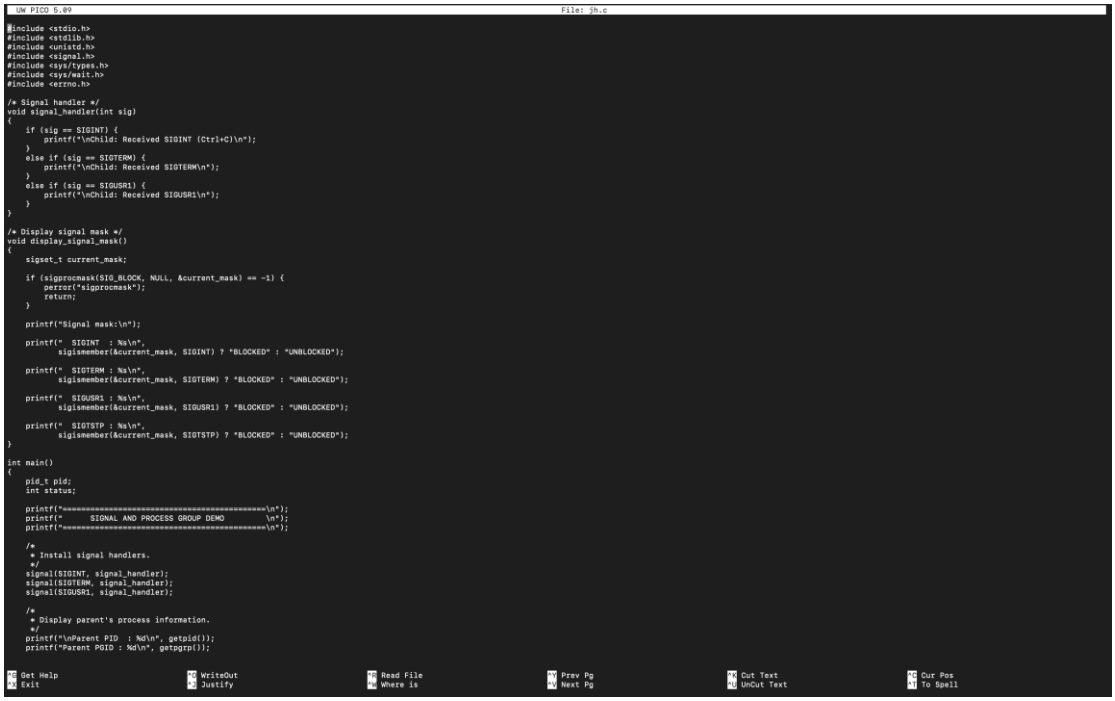
Task-2: 1.To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching



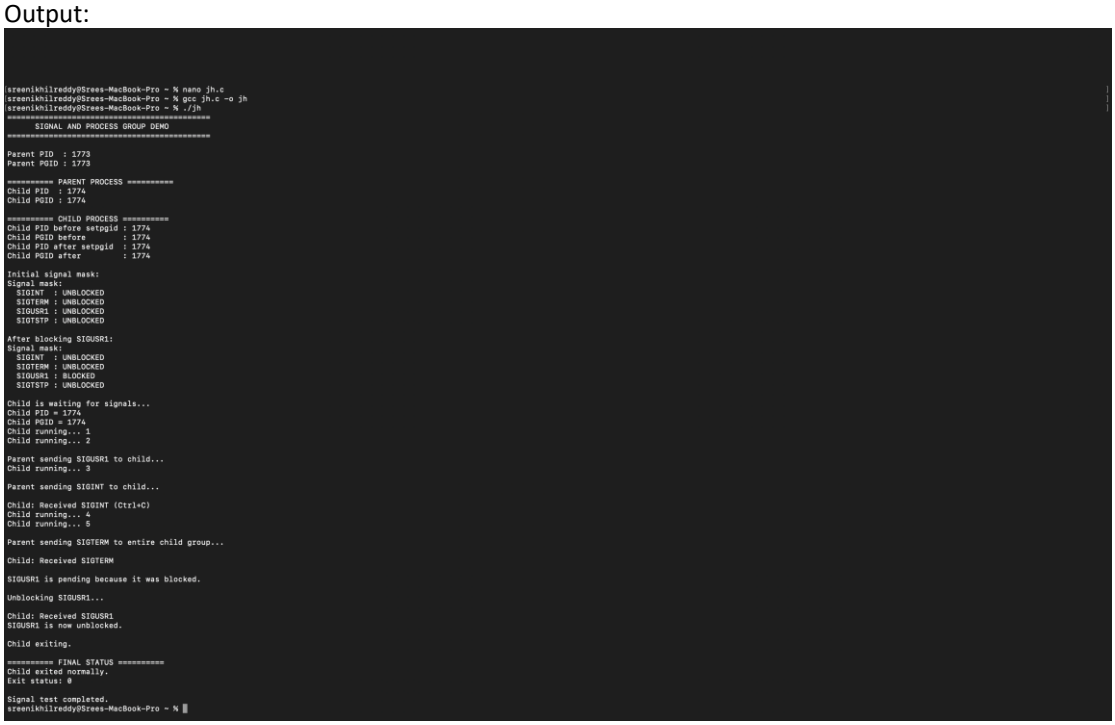
Output:



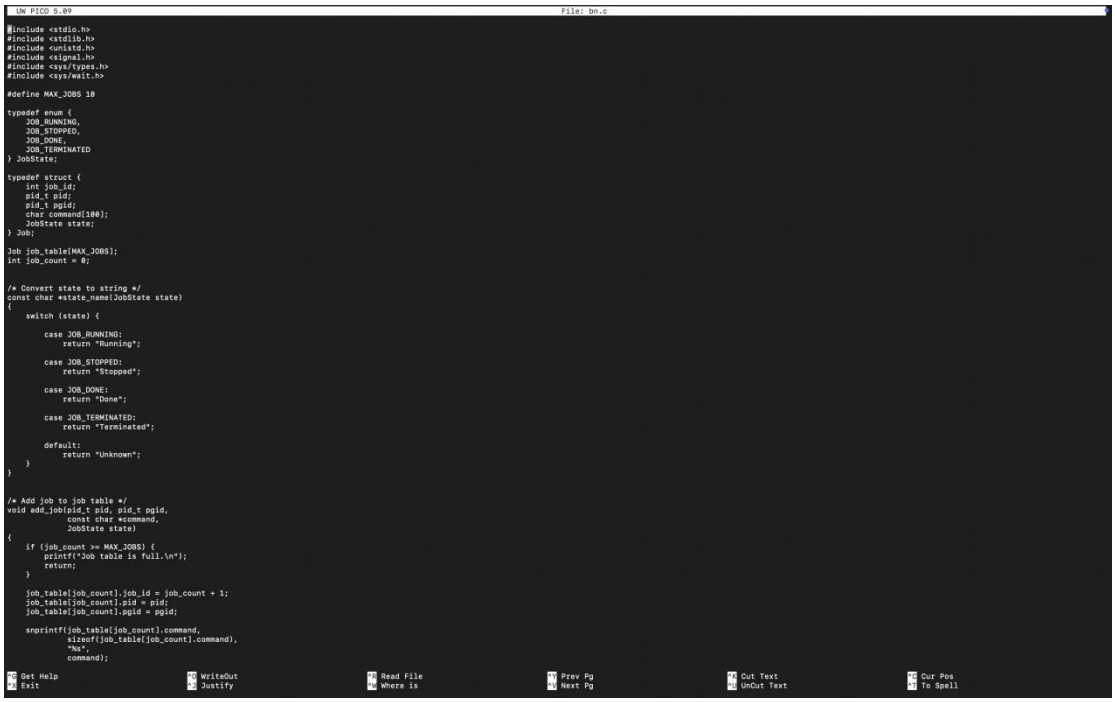
Task 3-: To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior

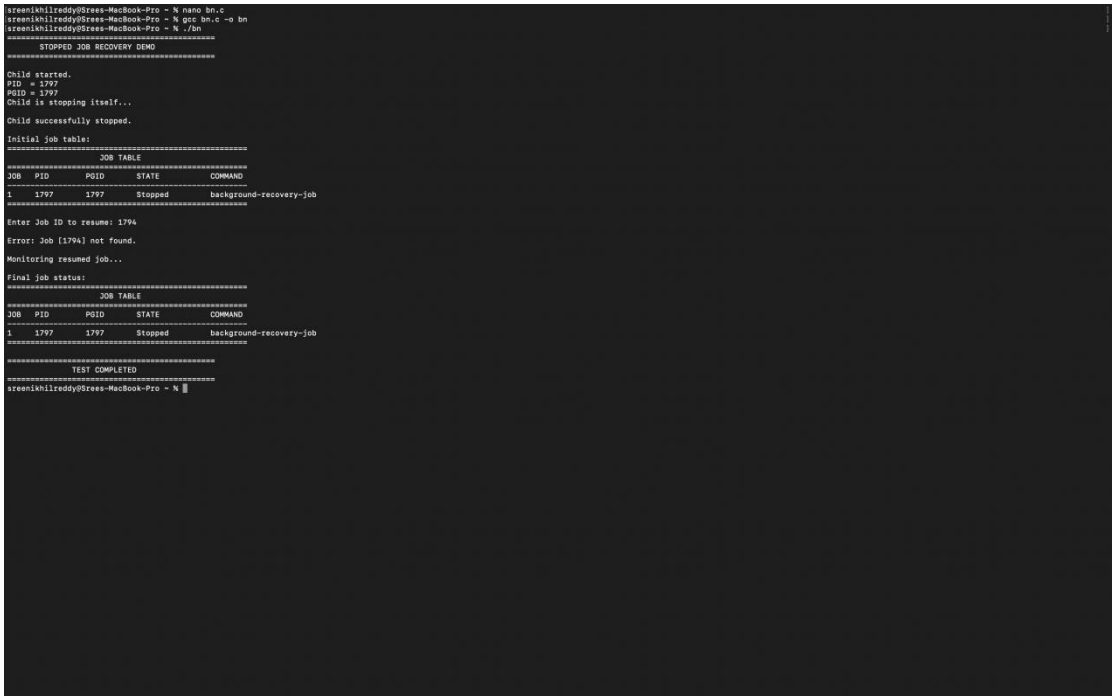


Output:



Task 4-: To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery




sreenikhilreddy@Srees-MacBook-Pro ~ % nano bn.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc bn.c -o bn
sreenikhilreddy@Srees-MacBook-Pro ~ % ./bn
STOPPED JOB RECOVERY DEMO
Child started.
PID = 1797
PGID = 1797
Child is stopping itself...
Child successfully stopped.
Initial job table:
JOB TABLE
JOB PID PGID STATE COMMAND
1 1797 1797 Stopped background-recovery-job
Enter Job ID to resume: 1794
Error: Job [1794] not found.
Monitoring resumed job...
Final job status:
JOB TABLE
JOB PID PGID STATE COMMAND
1 1797 1797 Stopped background-recovery-job
TEST COMPLETED
sreenikhilreddy@Srees-MacBook-Pro ~ %